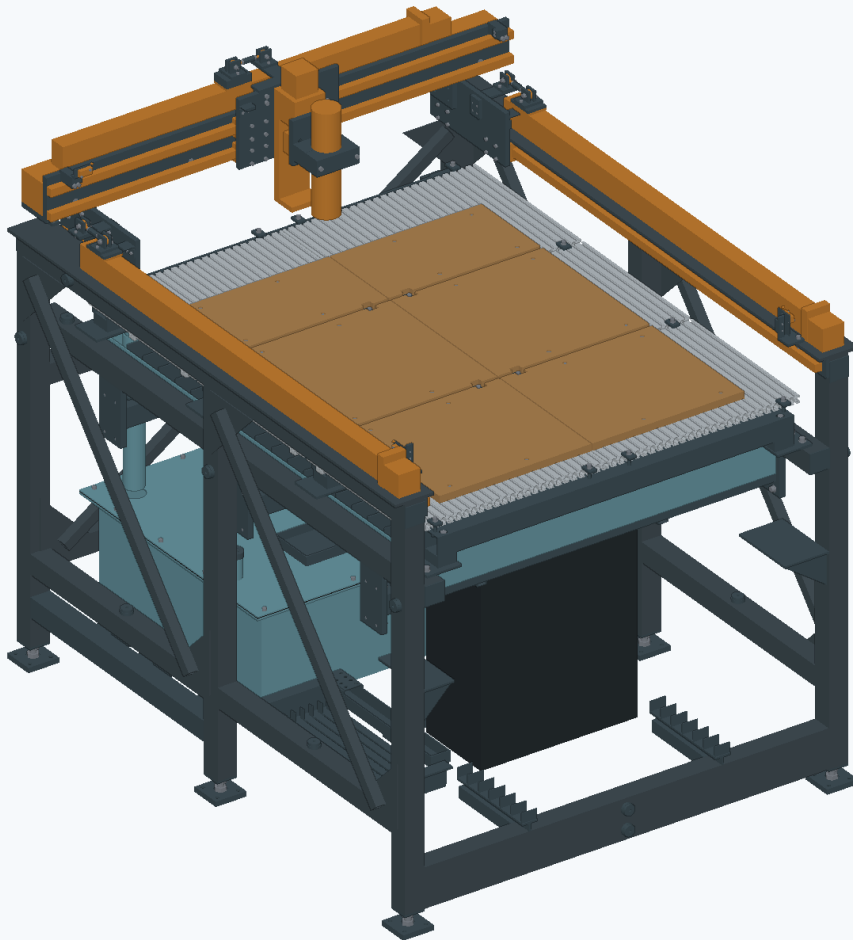


A machine built around the work.

Actual Rev G CAD: a steel chassis, a removable router bed and a fixed water pan. The design is corrected, with remaining interfaces identified.



ROUTER BED INSTALLED

The image is rendered from the exported CAD. Amber geometry represents purchased-part envelopes; appearance does not establish a verified supplier interface.

800 X / 1000 Y / 100 Z

Required nominal axis travel. The usable tool-and-workpiece envelope still depends on the final cutting tool, clamps and purchased motion interfaces.

A shorter path for cutting loads

The removable bed seats on dedicated supports in the steel chassis. The water pan remains a separate process component.

Parts stored in the stand

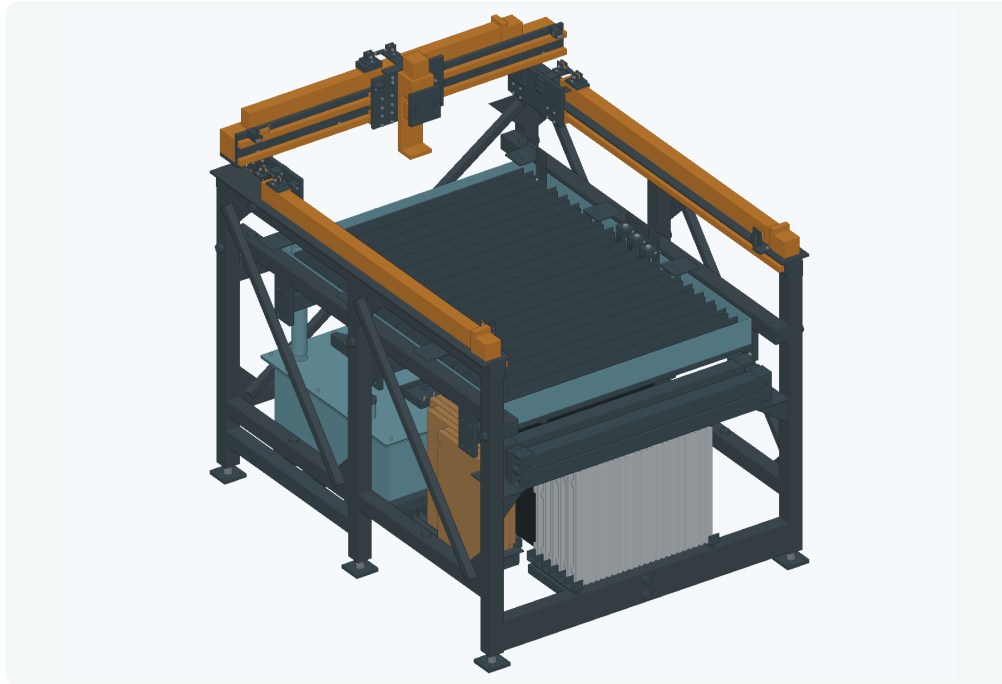
Six separate panels, four removable beams and separate spoilboards replace the earlier one-piece hoisted bed. The manual handling method is a working assumption pending your preference.

Review the design before ordering.

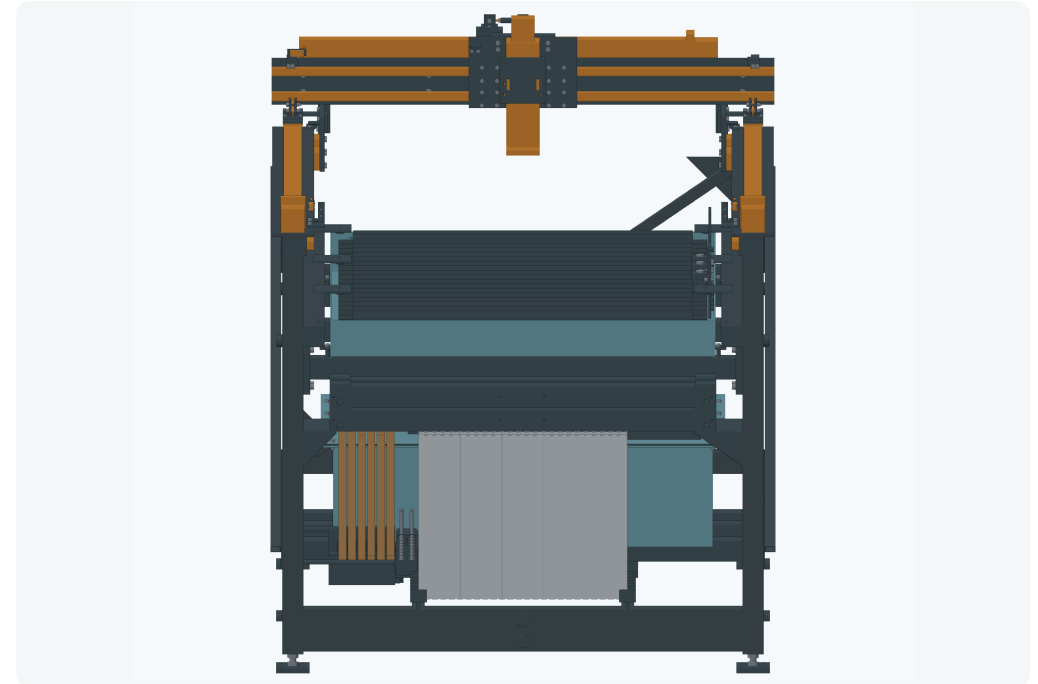
This brief replaces the previous hoisted-bed concept. It does not establish a finished machine or a revised build price.

Change the bed. Keep the footprint.

Manual conversion: 52 screws / clamp assemblies removed, plus 24 loose nuts retrieved. This is a budget-oriented design assumption, with no quick-change or automatic-handling claim.



STORED CONFIGURATION / ISOMETRIC



STORED CONFIGURATION / FRONT

Scope: bed and router tools are stored; a verified plasma torch, floating head and breakaway are still missing. Empty the front and left racks by reinstalling the bed before opening the controls cabinet.

01 / Boards and loose nuts

With the machine isolated and tool removed, take out 24 board screws. Store boards in order 4, 3, 2, 1, 6, 5. Retrieve all 24 loose top-slot nuts before rotating the panels.

02 / Panels and front seats

Remove 16 clamp assemblies and store six panels, rear slots first. The beam-1 parking pose on beam 2 permits removal of two front seats and their four screws; parking restraint remains unresolved.

03 / Beams and reinstallation

Release eight beam screws and store four beams in the specified order. Reverse to reinstall, remap the front seats and re-probe work zero. Physical restraint and hand access still require verification.

Geometry that can be checked.

Dimensions below describe the current model. Nominal shape checks, supplier measurements and structural qualification are separate pieces of evidence.

BED AND MATERIAL DEFINITION

Assembled deck	1003 x 1211 overall X x Y; includes panel gaps
Removable panels	6 at 500 x 397 x 33.775; panel + ties / hardware envelope
Panel dry mass	3.87 kg each estimated; weigh finished panels
Removable beams	4 at 924 long, 50.8 square x 3.048 wall tube; 5.06 kg estimated each
Working surfaces	Bare T-slot Z940.8; finished spoilboard Z958.8
Separate spoilboards	Six pieces: 19 rough to 18 finished. Only the reachable pieces are skimmed.
Removable retention	16 M6 clamps / 8 M8 beam screws / 24 M5 board screws / 4 M8 front-seat screws
Plasma slat support	Central 810 mm remains at Z850; 35 x 22 end reliefs clear the bed supports

Stock count: 5 two-packs = 10 bars. At three 397 mm lengths per bar, 1191 mm is assigned to finished pieces; 29 mm remains for saw kerfs and end trim. Confirm stock length and profile before cutting.

Drilled cabinet stringers

Two actual 6 mm bores now pass through both walls of each stringer. Their axes are local X12.7 and Y12.6 / 486.8. The screw-to-stringer collision exceptions were removed.

A defined drip shield

A two-piece welded steel roof and front lip, four cage tabs and four M5 fasteners replace the undefined bent sheet. The roof falls 1% toward the front. Its sampled withdrawal path clears the nominal enclosure.

Room below the drained-level guard

PAN_EMPTY has 10.37 mm nominal pan-floor clearance at its lowest slot-play position. The drawing requires at least 8 mm actual clearance after welding. Its real water trip and drain delay still need wet testing.

Matching adapter machining geometry

The two non-through adapter slots now retain their vertical orientation in DXF. The drawing-to-solid comparison passes. The purchased Z-carriage mating interface remains guarded.

What is verified. What closes the design.

Use the corrected CAD to resolve the remaining interfaces and load cases. A clean geometric model alone is not a fabrication release.

EVIDENCE IN THIS REVISION

Router state	1283 components; 0 unresolved static overlaps
Stored-bed state	1283 components; 0 unresolved static overlaps
STEP readback	Both state exports reimported successfully; solid counts and total volumes checked
Motion samples	9 full-machine router poses clear; corners and center, without actual cutter / cable envelopes
Large-part paths	36 panel, 25 beam and 60 spoilboard continuous segments checked, including previously stored parts
Drip shield service	24 forward positions + 11 lowering positions checked against the frame / enclosure reserve

Path clearance is not restraint. Stored parts and beam 1 in its temporary parking pose have no qualified anti-slide / anti-tip retention. Small hardware, front-seat and tool transfers, fingers, operator reach, cables and actual tooling remain outside these checks.

Current package: output/release-review/RevG-CAD
Repair evidence: output/cad-repair-2026-09-25
Project: github.com/DeadlySn777/plasma-router-stand

Purchased motion and tools

Verify actuator/guide interfaces, Z-carriage mounting and retention, tool projection and the owned torch. The Z drawing assigns 70 mm transverse spacing to the smaller 5 mm holes; thread and along-travel pitch remain unverified.

Rigidity and repeatability

Set a tool-to-work deflection target and cutting load cases. Qualify frame, beam, seat, fastener and foot behavior together; do not credit sand as added elastic stiffness.

Process and service hardware

Complete the selected drain valve, fittings, hoses, refill outlet, tank service path and enclosure door/cable/thermal arrangement. Verify wet operation and electrical interlocks using the actual equipment.

Handling, stock and cost

Confirm manual handling and design positive restraint. Twenty-four aluminum ties are required; one 12 x 12-inch plate does not supply them all. Verify owned quantity/alloy and recost the bill of materials. No new build total is asserted.